

Impact of liver-specific survival motor neuron (SMN) depletion on central nervous system and peripheral tissue pathology

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eLife Assessment

This work presents an **important** mouse model for a liver-specific depletion of the Survival Motor Neuron (SMN) protein, where the liver retains 30% of functional full-length SMN protein. The authors provide a profile of phenotypic changes in liver-specific SMN depleted mice with **convincing** evidence supporting their claims.

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Abstract

Spinal muscular atrophy (SMA) is an inherited neuromuscular disorder stemming from deletions or mutations in the Survival Motor Neuron 1 (*SMN1*) gene, leading to decreased levels of SMN protein, and subsequent motor neuron death and muscle atrophy. While traditionally viewed as a disorder predominantly affecting motor neurons, recent research suggests the involvement of various peripheral organs in SMA pathology. Notably, the liver has emerged as a significant focus due to the observed fatty liver phenotype and dysfunction in both SMA mouse models and SMA patients. Despite these findings, it remains unclear whether intrinsic depletion of SMN protein in the liver contributes to pathology in the peripheral or central nervous systems. To address this knowledge gap, we developed a mouse model with a liver-specific depletion of SMN by utilizing an *Alb-Cre* transgene together with one *Smn*^{2B} allele and one *Smn* exon 7 allele flanked by loxP sites. Initially, we evaluated phenotypic changes in these mice at postnatal day 19 (P19), a time when the severe model of SMA, the *Smn*^{2B/-} mice, typically exhibit many symptoms of the disease. Our findings indicate that liver-specific SMN depletion does not induce motor neuron death, neuromuscular pathology or muscle atrophy, characteristics typically observed in the *Smn*^{2B/-} mouse at P19. However, mild liver steatosis was observed at this time point, although no changes in liver function were detected. Notably, pancreatic alterations resembled that of *Smn*^{2B/-} mice, with a decrease in insulin-producing β -cells and an increase in glucagon-producing α -cells,

accompanied by a reduction in blood glucose and an increase in plasma glucagon and glucagon-like peptide (GLP-1) levels. Moreover, these changes were transient, as P60 mice exhibited recovery of liver and pancreatic function. While the mosaic pattern of the Cre-mediated excision precludes definitive conclusions regarding the contribution of liver-specific SMN depletion to overall tissue pathology, our findings highlight an intricate connection between liver function and pancreatic abnormalities in SMA, adding a nuanced layer to our understanding of the disease's complexities.

1. Introduction

Insufficient levels of Survival Motor Neuron (SMN) protein, primarily arising from mutations or deletions in the *SMN1* gene, are the root cause of spinal muscular atrophy (SMA), a hereditary neuromuscular disorder widely acknowledged as one of the leading genetic causes of infant mortality^{1,2,3}. Characterized by the loss of motor neurons, SMA presents as a progressive weakening of muscle strength, primarily impacting proximal muscles. If left untreated, this deterioration culminates in respiratory failure and premature mortality⁴.

The diverse phenotypes observed in SMA patients can be attributed to the genetic variations within the *SMN* gene¹. Humans harbor two copies of the *SMN* gene, *SMN1* and *SMN2*, which share a nearly identical sequence except for a critical C to T substitution at position 6 of exon 7 in *SMN2*⁵. This single nucleotide change disrupts splicing of the gene, resulting in the predominant production of an unstable protein lacking exon 7, known as *SMNΔ7*⁵. Despite this, approximately 10% of functional full-length (FL) SMN protein is still produced by the *SMN2* gene². Consequently, the *SMN2* gene serves as a modifier of disease phenotype, with a higher copy number correlating with milder disease severity⁶. In contrast, mice possess a single copy of the *Smn* gene². Notably, while homozygous deletion of both *SMN* genes has not been observed in humans, the homozygous deletion of the *Smn* gene in mice induces morphological alterations and degenerative changes in embryos post-morula stage, ultimately resulting in death at preimplantation stage⁷. This underscores the pivotal role of the *SMN* gene in early embryonic development⁸.

Recent research has unveiled the extensive impact of SMN depletion beyond motor neurons, implicating additional tissues such as the liver and pancreas in both SMA patients and mouse models of SMA^{9–12}. Despite major therapeutic advancements focusing on restoring SMN levels, which have notably enhanced both lifespan and quality of life in SMA patients, complete phenotypic rescue remains elusive^{4,13,14}. Additionally, existing therapies primarily target postnatal treatment, potentially overlooking crucial developmental changes stemming from SMN depletion. This underscores the likelihood of a broader role for the *SMN* gene in multiple non-neuronal cell types, necessitating further exploration.

Attention has been increasingly drawn to the liver as a critical area of investigation in SMA research. Both clinical studies and preclinical mouse models have unveiled disruptions in fatty acid metabolism, leading to an increased susceptibility among SMA patients to dyslipidemia and liver steatosis^{12,15–18}. Given its multifaceted role in regulating numerous biological functions, including the storage and regulation of lipids, carbohydrates, amino acids, and iron, as well as the synthesis of essential growth factors and the clearance of toxic metabolites, any disruptions in liver function may have deleterious effects on other tissues¹⁹.

To explore the role of SMN in the liver, prior studies have utilized a transgenic mouse model featuring the Cre recombinase transgene driven by the α -fetoprotein promoter (*Alfp-Cre*). This approach led to near-total depletion of SMN protein specifically in the liver, resulting in severe impairment of liver development and late embryonic lethality⁸. In our study, to better represent the level of SMN depletion in the liver observed in SMA and elucidate the intricate interplay

between SMN deficiency in the liver and its systemic impact in SMA-like pathology, we utilized a mouse model employing *Alb-Cre*, where the albumin promoter drives expression of Cre recombinase in a liver-specific manner. This model also harbors one *Smn*^{2B} allele and one *Smn* exon 7 allele flanked by loxP sites (F7). The *Smn*^{2B} allele contains a 3-base pair substitution in the mouse *Smn* gene, resulting in aberrant splicing of exon 7^{20,21}. Consequently, this genetic approach results in the production of approximately 30% full-length functional SMN protein production in the liver.

Our findings reveal that our novel liver-specific SMN depleted (herein referred to as *Alb*^{Cre/+};*Smn*^{2B/F7}) mouse model induces mild liver steatosis and disrupts pancreatic function at postnatal day 19 (P19), characterized by decreased insulin-positive and increased glucagon-positive cells, alongside reduced blood glucose levels. However, both liver and pancreatic defects were rescued by P60. Remarkably, this mouse model does not exhibit an overt SMA phenotype, demonstrating normal lifespan and motor function. Nevertheless, our data suggests an intricate relationship between hepatic function and pancreatic abnormalities in SMA, thereby enhancing our comprehension of the disease's pathophysiology.

2. Results

2.1 Generation of liver-specific SMN-depleted mice

To elucidate the impact of liver-specific SMN depletion on both CNS and peripheral tissue pathology, we utilized mice expressing Cre recombinase under the control of the mouse albumin promoter, referred to as *Alb*^{Cre}. This strain is commonly employed for liver-specific gene knockout using the Cre/loxP system²². We generated the *Smn*^{2B/F7} genotype by crossing *Smn*^{F7/F7} with *Smn*^{2B/2B} mice (Fig. 1A). The *Smn*^{2B/F7} genotype carries one allele harboring a three nucleotide switch in the exon splicing enhancer within exon 7 of the mouse *Smn* gene (known as *Smn*^{2B} allele), mimicking the SMN2 gene in humans²¹, while the other allele features exon 7 flanked by loxP sites²³. Additionally, crosses between *Alb*^{Cre} and *Smn*^{F7/F7} yielded *Alb*^{Cre/+};*Smn*^{F7/+} offspring.

The liver-specific mouse line was subsequently established through further genetic crosses between *Alb*^{Cre/+};*Smn*^{F7/+} and *Smn*^{2B/F7}, resulting in various genotypes, including +/+;*Smn*^{2B/F7} (Fig. 1B), and *Alb*^{Cre/+};*Smn*^{2B/F7} (Fig. 1C). As +/+;*Smn*^{2B/F7} mice lack Cre recombinase, they were used as heterozygous controls. In contrast, recombination occurs in *Alb*^{Cre/+};*Smn*^{2B/F7} animals resulting in the specific deletion of one allele of exon 7 in the liver, while the *Smn*^{2B} allele remains unaffected.

Smn^{2B/-} mice, the severe mouse model of SMA, express approximately 15% residual full-length SMN (FLSMN) compared to *Smn*^{2B/+} heterozygous mice²⁴. Therefore, both genotypes were included in this study as additional controls. Tissues were collected at postnatal day 19 (P19), corresponding to the symptomatic phase of the *Smn*^{2B/-} mouse model. At this stage, *Smn*^{2B/-} mice show increased liver steatosis and metabolic defects, ultimately culminating in multiple system defects and mortality, typically occurring around P25²⁵. In contrast, *Smn*^{2B/+} mice have slightly reduced levels of FLSMN and no overt phenotypical changes compared to wild-type mice¹⁰.

We first assessed SMN protein levels in the liver and across CNS and peripheral tissues. Our findings confirmed a reduction in SMN protein levels across various tissues in *Smn*^{2B/-} mice (Fig. 1D-H). However, in *Alb*^{Cre/+};*Smn*^{2B/F7}, there was a ~70% decrease in FLSMN protein expression in the liver (Fig. 1D, D') compared to littermate control counterparts (+/+;*Smn*^{2B/F7}). Conversely, SMN protein levels in the brain (Fig. 1E, E'), muscle (Fig. 1F, F'), spinal cord (Fig. 1G, G') and pancreas (Fig. 1H, H') remained unaltered, indicating that deletion of *Smn* exon 7 was restricted to the liver in *Alb*^{Cre/+};*Smn*^{2B/F7} mice. We validated these findings by assessing Cre expression in liver, pancreas, and spinal cord tissues (Supp. Fig. 1A). Cre expression was

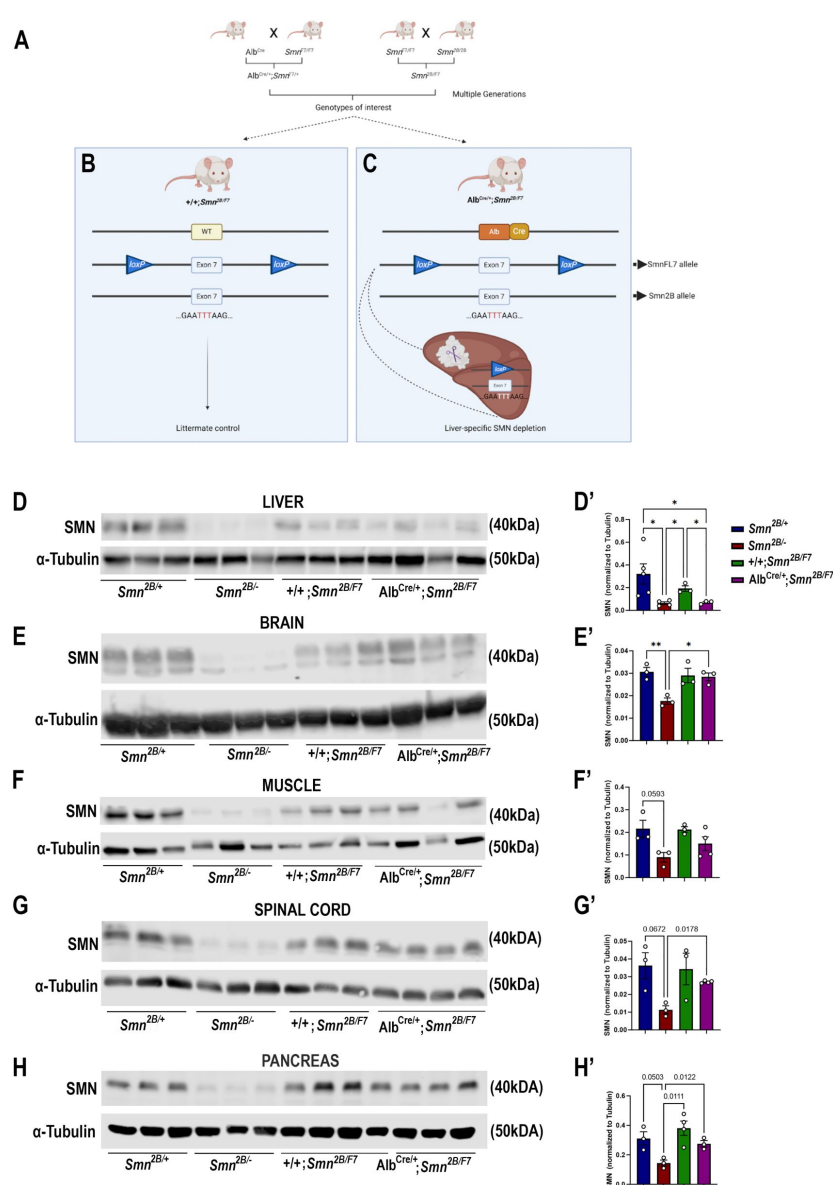


Figure 1.

Generation of liver-specific SMN-depleted mice.

A. Schematic representation of the breeding scheme to generate liver-specific SMN depletion. In these mice, the *Smn* gene carries one *Smn*^{2B} allele that harbors a three-nucleotide switch in exon 7, and one *Smn* allele whose exon 7 is floxed by loxP sites (*Smn*^{FL7}). The *Alb-Cre* transgene has also been crossed into this background to provide liver-specific expression of the Cre recombinase. B. Littermate controls do not harbor Cre recombinase but do carry one *Smn*^{2B} allele and are thus used as heterozygous controls. C. Upon crossing *Alb*^{Cre/+}; *Smn*^{FL7/+} with *Smn*^{2B/+} animals, Cre recombination will induce specific excision of exon 7 in the liver, leading to reduced SMN protein production only in the liver. D-H. Immunoblots were performed to assess SMN protein levels in various tissues from *Smn*^{2B/+}, *Smn*^{2B/-}, +/+; *Smn*^{2B/FL7}, and *Alb*^{Cre/+}; *Smn*^{2B/FL7} mice at P19. Membranes were probed for SMN and then reprobed for α-tubulin (loading control). D'-H'. Bar graphs show quantification of SMN protein levels in the liver (D'), brain (E'), muscle (F'), spinal cord (G') and pancreas (H'), normalized to α-tubulin. nD≥D3, mean ± SEM. Statistical significance indicated by *pD<D0.05, **pD<D0.01 or p values, following Brown-Forsythe and Welch ANOVA.

exclusively observed in the liver, where it exhibited variable expression across hepatocytes, with some cells showing detectable expression while others showed no expression, confirming the stochastic nature of Cre activity in our model. No Cre signal was detected in either the pancreas or spinal cord tissues.

2.2 $Alb^{Cre/+};Smn^{2B/F7}$ mice show mild yet important liver steatosis

We previously showed that $Smn^{2B/-}$ mice exhibit defects in fatty acid metabolism and lipid accumulation in the liver^{12,16,18}. Additionally, SMA patients face an elevated risk of dyslipidemia and steatosis¹². Hence, we explored whether the intrinsic decrease in SMN levels restricted to the liver would lead to increased lipid accumulation. Histology from $Alb^{Cre/+};Smn^{2B/F7}$ mice (**Fig. 2A**) qualitatively indicates some level of hepatic microvesicular steatosis and lipid accumulation. However, there was a noticeable variation in cell morphology and Oil Red-O retention among hepatocytes, suggesting incomplete recombination.

Consistent with the histological findings, $Alb^{Cre/+};Smn^{2B/F7}$ mice showed variable triglycerides levels ($n_1=45.95$, $n_2=10.10$; $n_3=75.72$; $n_4=21.91$ $\mu\text{g}/\text{mg}$, mean $\pm$ SEM 38.44 ± 14.48) (**Fig. 2B**), and yet higher than those observed in $+/+;Smn^{2B/F7}$ controls ($n_1=2.60$, $n_2=0.15$; $n_3=0.66$; $n_4=1.44$ $\mu\text{g}/\text{mg}$, mean $\pm$ SEM 1.213 ± 0.5332), although this difference did not reach statistical significance ($p=0.0824$). As expected, $Smn^{2B/-}$ controls exhibited increased lipid accumulation and triglyceride levels. Taken together, the data suggest an intrinsic role of SMN in regulating fat liver metabolism.

We next sought to identify whether $Alb^{Cre/+};Smn^{2B/F7}$ mice display disturbed liver homeostatic function. Specifically, we evaluated parameters associated with iron metabolism (**Fig. 2C,C',D,D'**), endoplasmic reticulum stress (**Fig. 2E,E'**), and insulin-like growth factor-1 (IGF-1) (**Fig. 2F**) levels, all of which have previously been identified as altered in $Smn^{2B/-}$ mice¹⁸ and were further confirmed in our study. Our analysis revealed no significant differences in the protein levels of heme oxygenase, transferrin, P62, and IGF-1 between the $Alb^{Cre/+};Smn^{2B/F7}$ and $+/+;Smn^{2B/F7}$ control groups. These findings suggest that liver-specific SMN depletion may not play a significant role in these pathways or that the mosaic nature of Cre-mediated excision led to a subtle effect that is difficult to detect.

2.3 Liver-specific SMN-depleted mice display pancreatic defects

We have demonstrated pancreatic and glucose abnormalities in the $Smn^{2B/-}$ mouse model of SMA and type I SMA patients^{11,26}. Therefore, we conducted immunohistochemistry to investigate the cellular profile of the pancreas, aiming to ascertain whether the targeted depletion of SMN in the liver contributes to pancreatic dysfunction. Surprisingly, double-labeling of insulin-producing β cells and glucagon-producing α cells revealed a significant decrease in the proportion of β cells and a notable increase in α cells in the pancreas of $Alb^{Cre/+};Smn^{2B/F7}$ mice compared to $+/+;Smn^{2B/F7}$ and $Smn^{2B/+}$ heterozygous controls (**Fig. 3A,B,F**). This loss of β cells and rise in α cells could suggest a functional impairment in glucose metabolism. Consequently, we evaluated non-fasting glucose levels in these mice (**Fig. 3C**), revealing reduced blood glucose levels in $Alb^{Cre/+};Smn^{2B/F7}$ mice relative to $+/+;Smn^{2B/F7}$ and $Smn^{2B/+}$. However, blood glucose levels remained higher than those observed in $Smn^{2B/-}$ mice. Overall, these findings indicate imbalances in α and β cell fate within pancreatic islets in $Alb^{Cre/+};Smn^{2B/F7}$ mice, resembling those observed in $Smn^{2B/-}$ mice.

In previous work, we demonstrated that $Smn^{2B/-}$ mice exhibit a normoinsulinemic plasma state, despite persistent hypoglycemia and reduced C-peptide levels at P19¹⁸, findings that were recapitulated in this study (**Fig. 3C-E**). Similarly, $Alb^{Cre/+};Smn^{2B/F7}$ mice showed no significant changes in plasma insulin levels. However, surprisingly, C-peptide levels in these mice were comparable to those observed in $+/+;Smn^{2B/F7}$ and $Smn^{2B/+}$ heterozygous controls (**Fig. 3E**). Additionally, we observed a marked increase in plasma glucagon levels (**Fig. 3G**) as well as elevated glucagon-like peptide (GLP-1) levels, a product of proglucagon processing (**Fig. 3H**),

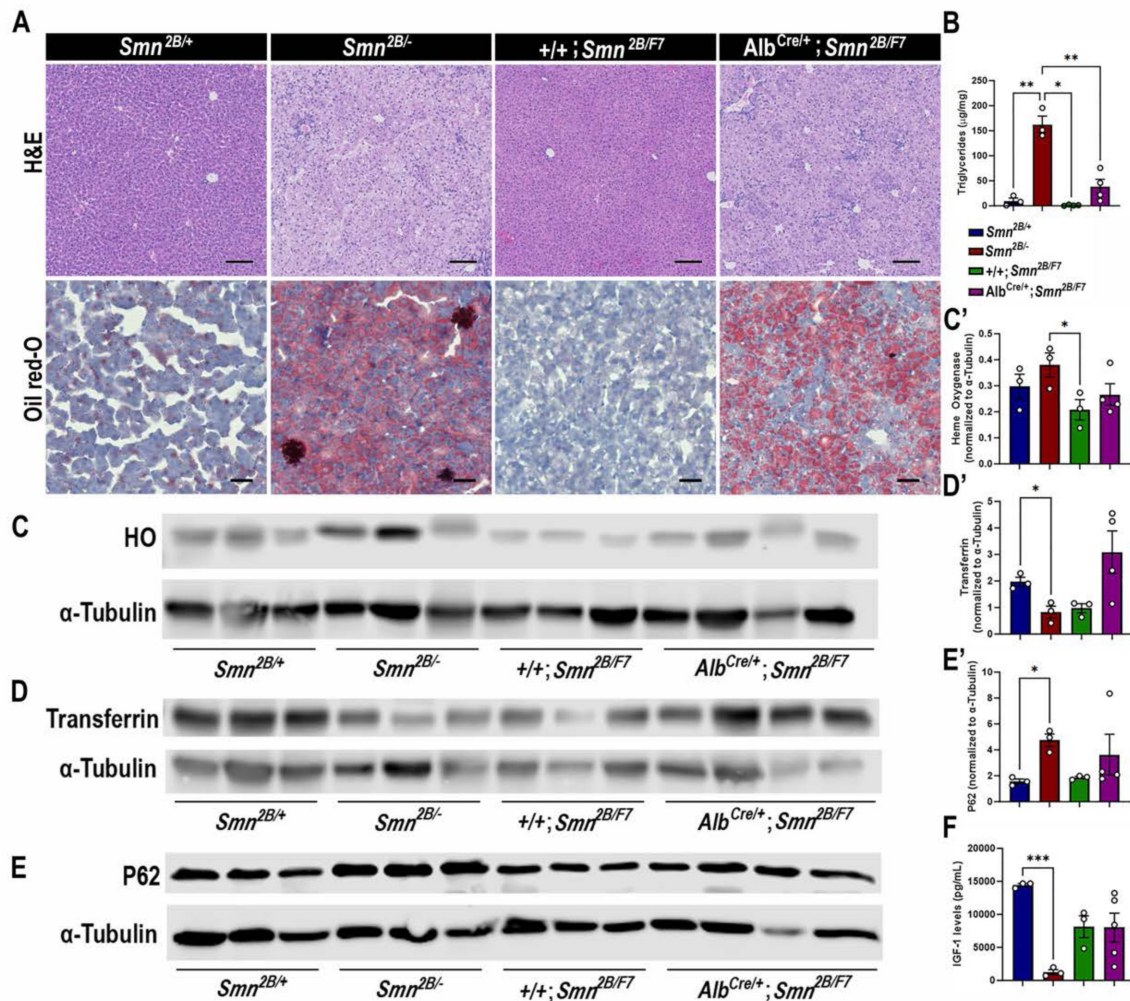


Figure 2.

Impact of liver-specific SMN depletion in liver function.

A. Representative images of H&E (top row, scale bar 100 μm) and Oil Red-O (bottom row, scale bar 50 μm) stained liver sections from *Smn*^{2B/+}, *Smn*^{2B/-}, *+/+; Smn*^{2B/F7}, and *Alb*^{Cre/+}; *Smn*^{2B/F7} mice at P19. B. Bar graph shows quantification of liver triglycerides. C-E. Immunoblots were performed to assess heme oxygenase (HO, C-C'), transferrin (D-D') and P62 (E-E') protein levels in the liver. α-tubulin was used as a loading control. C'-E' Bar graphs show quantification of HO (C'), transferrin (D'), and P62 (E') levels. F. Bar graph depicts quantification of liver IGF-1. nD≥3, mean ± SEM. Statistical significance indicated by **p*D<0.05, ***p*D<0.01, ****p*D<0.001, following Brown-Forsythe and Welch ANOVA.

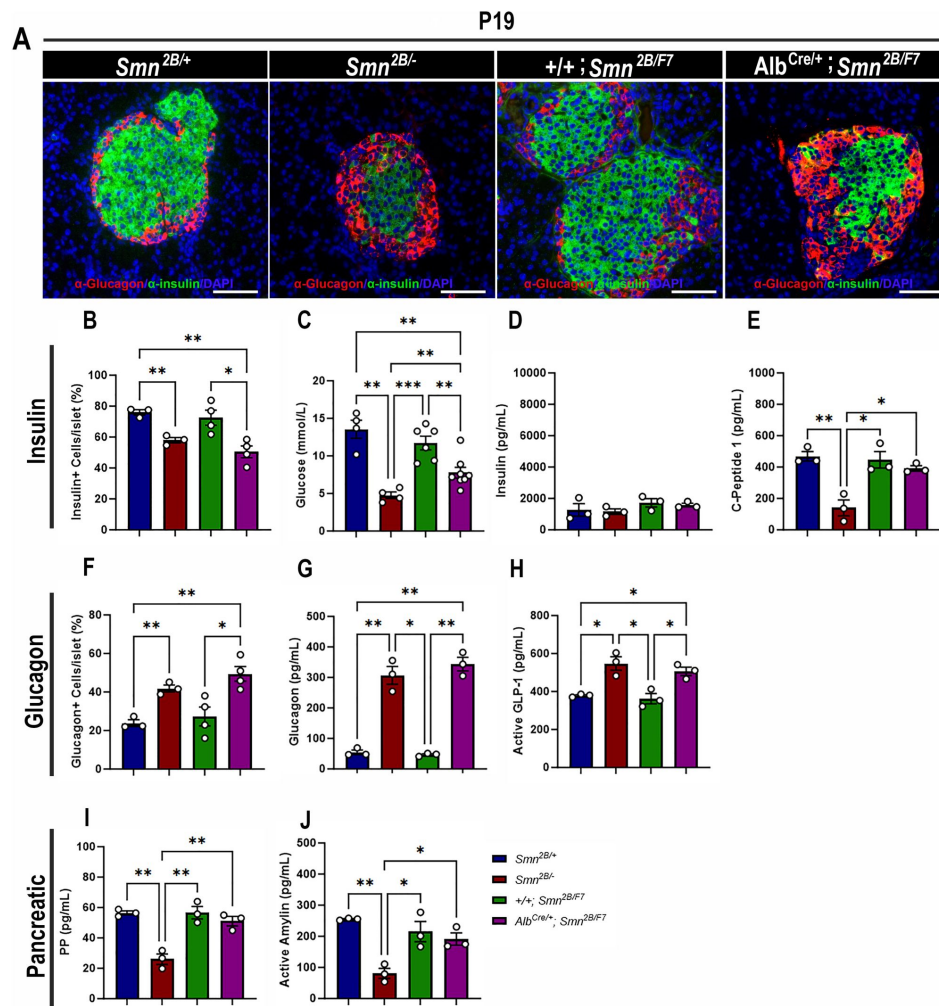


Figure 3.

Contribution of liver-specific SMN depletion to pancreatic pathology.

A. Representative immunofluorescent images of pancreatic islets stained for glucagon (red) and insulin (green) from $Smn^{2B/+}$, $Smn^{2B/-}$, $+/+; Smn^{2B/F7}$, and $Alb^{Cre/+}; Smn^{2B/F7}$ mice at P19. Scale bar 50 μ m. B. Bar graph shows quantification of insulin-positive cells relative to the total number of DAPI-positive cells within the pancreatic islet. C. Bar graph depicts non-fasting blood glucose levels across different genotypes. D-E. Bar graphs illustrate non-fasting plasma levels of insulin (D) and C-peptide (E). F. Bar graph shows quantification of glucagon-positive cells relative to the total number of DAPI-positive cells within the pancreatic islet. G-J. Bar graphs depict non-fasting plasma levels of glucagon (G), active GLP-1 (H), pancreatic polypeptide (I) and active amylin (J). $n \geq 3$, mean $\pm$ SEM. Statistical significance indicated by * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, following Brown-Forsythe and Welch ANOVA.

reaching levels similar to those seen in *Smn*^{2B/-} mice. These changes likely reflect an increase in pancreatic α -cell numbers, though they were not accompanied by dysregulation of plasma pancreatic polypeptide (PP, **Fig. 3I**) or amylin (**Fig. 3J**) hormone production. Furthermore, the maintenance of SMN protein expression at levels comparable to those of heterozygous controls in the pancreas (**Fig. 1H,H'**) suggests that selective SMN depletion in the liver may be implicated in the dysfunction of the pancreas-liver axis.

2.4 Motor neurons, neuromuscular junction (NMJ) or motor fibers are unaffected in liver-specific SMN-depleted mice

The hallmark features of SMA include the degeneration of lower spinal motor neurons, neuromuscular junction (NMJ) pathology, and muscle atrophy^{10,25,27,33}. While the exact mechanisms linking these changes in the motor unit to the clinical manifestations of SMA remain unclear, diminished levels of SMN protein are recognized as a pivotal factor^{1,2,25,33,35}. Thus, we investigated the impact of liver-specific SMN depletion on these SMA-like features. Our findings show no significant differences in motor neuron cell body counts (**Fig. 4A,B**), neurofilament accumulation at NMJs (**Fig. 5A,B**), NMJ endplate occupancy (**Fig. 5A,C**), muscle fiber size (**Fig. 6A,B**), or muscle fiber frequency distribution (**Fig. 6C**) among *Alb*^{Cre/+};*Smn*^{2B/F7} and +/+;*Smn*^{2B/F7} and *Smn*^{2B/+} heterozygous controls. However, as anticipated, *Smn*^{2B/-} mice exhibited motor neuron loss, NMJ defects, and muscle atrophy.

2.5 Liver-specific SMN-depleted mice show transitory liver and pancreatic pathology

We next assessed *Alb*^{Cre/+};*Smn*^{2B/F7} mice at P60 to determine whether liver and pancreatic dysfunction persisted or improved during early adulthood. Histological analysis revealed no significant changes in hepatocyte morphology (as assessed by H&E staining) or lipid accumulation (evaluated with Oil Red O staining) in *Alb*^{Cre/+};*Smn*^{2B/F7} mice at this time point (**Fig. 7A**). These findings suggest that proliferating non-recombined cells may help restore liver homeostasis over time. Notably, the expression of Cre recombinase was also observed at P60 (Supp. Fig. 1B), indicating stable Cre activity across postnatal development.

Further examination of pancreatic tissue at P60 showed no significant differences in insulin-producing β cells or glucagon-producing α cells in *Alb*^{Cre/+};*Smn*^{2B/F7} mice compared to +/+;*Smn*^{2B/F7} and *Smn*^{2B/+} heterozygous controls (**Fig. 7B, C, G**). Additionally, non-fasting plasma glucose, insulin, and C-peptide levels in *Alb*^{Cre/+};*Smn*^{2B/F7} mice were similar to those in the control groups (**Fig. 7D-F**). Moreover, no significant changes in plasma glucagon (**Fig. 7H**), GLP-1 (**Fig. 7I**), pancreatic polypeptide (**Fig. 7J**), or amylin hormone production were observed. Collectively, these results suggest that the liver and pancreatic defects observed in *Alb*^{Cre/+};*Smn*^{2B/F7} mice may be transient, potentially linked to the stochastic nature of Cre recombinase activity in our model and the highly proliferative properties of hepatocytes. In addition, these findings suggest that liver-specific SMN may play a role in regulating pancreatic function, a mechanism that warrants further investigation.

2.6 Liver-specific SMN depletion does not affect major adipocyte and gastrointestinal hormones

Gastrointestinal (GI) hormones play a key role in regulating hepatic lipid metabolism and pancreatic function³⁶. Thus, to assess whether adipocytes and GI hormones contribute to liver and pancreatic dysfunction in liver-specific SMN-depleted mice, we measured plasma levels of major adipocyte hormones (leptin and resistin) and GI hormones (ghrelin, gastric inhibitory polypeptide (GIP), peptide YY (PYY) and secretin) at P19 and P60^{36,37}. Our data revealed no significant changes in adipocyte hormones at either time point (**Fig. 8A-B**, G-H) across all

Figure 4.

Impact of liver-specific SMN depletion on motor neuron cell body numbers.

A. Representative images of lumbar spinal cord anterior horns stained for ChAT (red) and DAPI (blue) from *Smn*^{2B/+}, *Smn*^{2B/-}, *+/+; Smn*^{2B/F7}, and *Alb*^{Cre/+}; *Smn*^{2B/F7} mice at P19. B. Bar graph shows quantification of motor neuron cell body numbers. $n \geq 3$, mean $\pm$ SEM. Statistical significance indicated by $**p < 0.01$, $***p < 0.001$, following Brown-Forsythe and Welch ANOVA. Scale bar = 50 μ m.

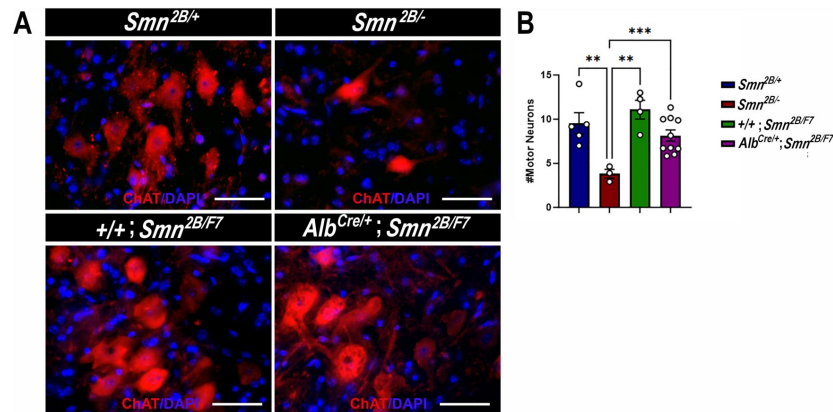
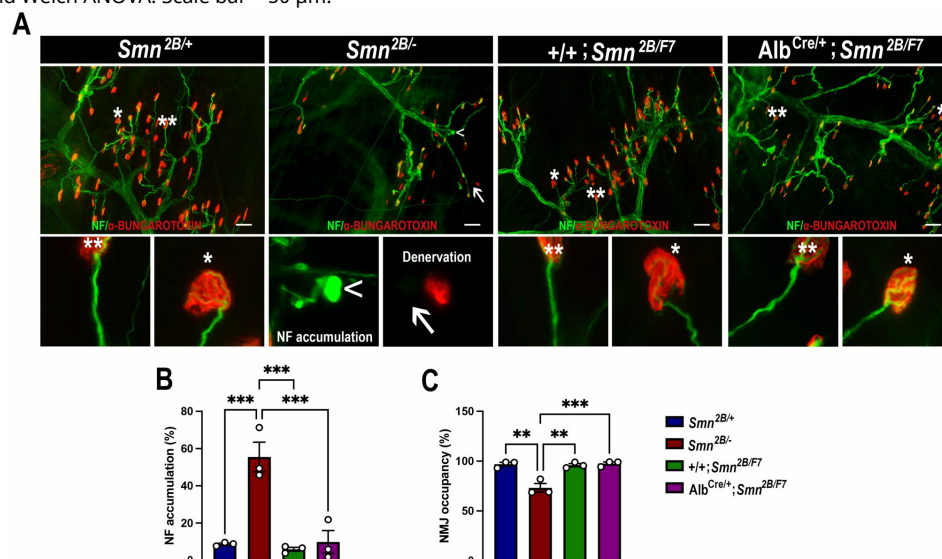


Figure 5.

Impact of liver-specific SMN depletion on neuromuscular junction pathology.

A. Representative images of *transversus abdominis* (TVA) muscle stained with bungarotoxin (red), neurofilament (NF) (green) and synaptic vesicle protein 2 (green) from *Smn*^{2B/+}, *Smn*^{2B/-}, *+/+; Smn*^{2B/F7}, and *Alb*^{Cre/+}; *Smn*^{2B/F7} mice at P19. *Depicts occupied endplate, whereas **depicts normal neurofilament distribution; arrowheads show NF accumulation and arrows show unoccupied endplate/denervation. B-C. Bar graphs show quantification of neurofilament accumulation (B) and endplate occupancy (C). $n = 3$, mean $\pm$ SEM. Statistical significance indicated by $**p < 0.01$, $***p < 0.001$, following Brown-Forsythe and Welch ANOVA. Scale bar = 50 μ m.



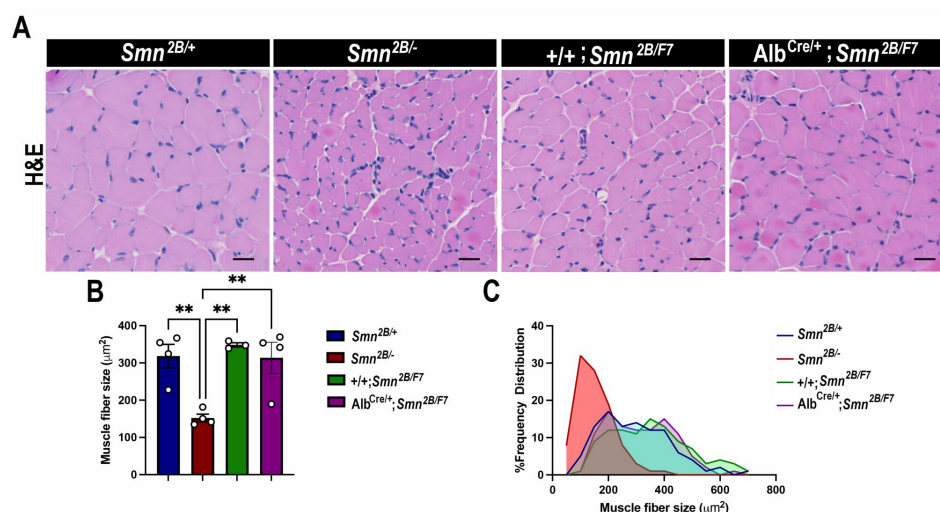


Figure 6.

Impact of liver-specific SMN depletion on muscle fiber size and distribution.

A. Representative images of H&E-stained *tibialis anterior* muscle sections from *Smn*^{2B/+}, *Smn*^{2B/-}, +/+;*Smn*^{2B/F7}, and *Alb*^{Cre/+};*Smn*^{2B/F7} mice at P19. B. Bar graph shows quantification of muscle fiber cross-sectional area. C. Graph demonstrating frequency of muscle fiber size across different genotypes. n ≥ 3, mean ± SEM. Statistical significance indicated by **p < 0.01, following Brown-Forsythe and Welch ANOVA. Scale bar = 20 μm.

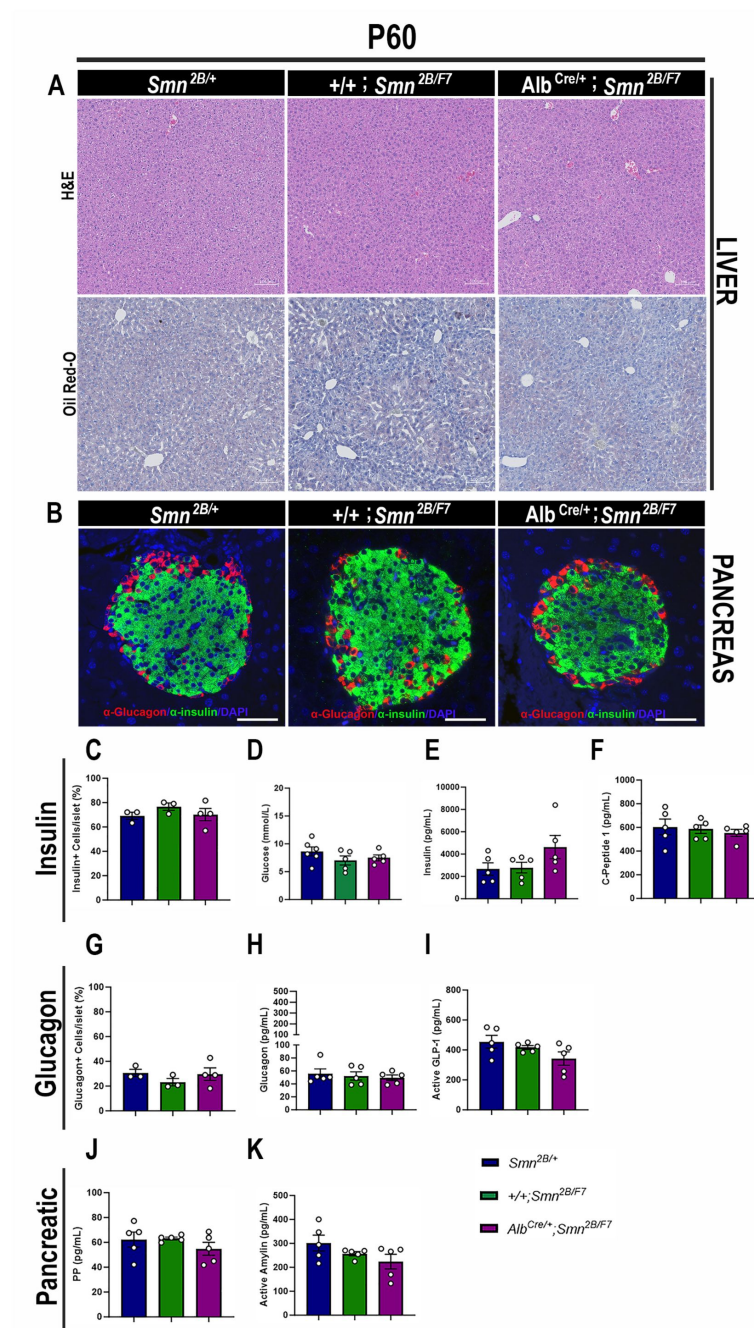


Figure 7.

Consequences of liver-specific SMN depletion to early adulthood liver and pancreatic pathology.

A. Representative images of H&E (top row) and Oil Red-O (bottom row) stained liver sections from *Smn*^{2B/+}, *+/+ ; Smn*^{2B/F7}, and *Alb*^{Cre/+} ; *Smn*^{2B/F7} mice at P60. Scale bar 100 μ m. B. Representative immunofluorescent images of pancreatic islets stained for glucagon (red) and insulin (green) from *Smn*^{2B/+}, *+/+ ; Smn*^{2B/F7}, and *Alb*^{Cre/+} ; *Smn*^{2B/F7} mice at P60. Scale bar 50 μ m. C. Bar graph shows the quantification of insulin-positive cells relative to the total number of DAPI-positive cells within the pancreatic islet. D. Bar graph depicts non-fasting blood glucose levels from P60 mice across different genotypes. E-F. Bar graphs illustrate non-fasting plasma levels of insulin (E) and C-peptide (F). G. Bar graph shows quantification of glucagon-positive cells relative to the total number of DAPI-positive cells within the pancreatic islet. H-K. Bar graphs depict non-fasting plasma levels of glucagon (H), active GLP-1 (I), pancreatic polypeptide (J) and active amylin (K). $n \geq 3$, mean $\pm$ SEM. Samples were analyzed using Brown-Forsythe and Welch ANOVA, and no statistically significant differences were observed.

genotypes. Moreover, few alterations in GI hormones (**Fig. 8C-F**, I-L) were observed exclusively in *Smn*^{2B/-} mice at P19, including a marked reduction of ghrelin (**Fig. 8C**) – a hormone involved in glucose homeostasis in pancreatic islet and liver³⁸ – and secretin (**Fig. 8F**), which regulates secretory activity in organs such as the liver and pancreas³⁹. Notably, no such changes were observed in *Alb*^{Cre/+};*Smn*^{2B/F7} at any time point. Altogether, these results suggest that the pancreatic and liver changes at P19 are likely independent of major adipocyte and GI hormone alterations, pointing to a more direct disruption of the liver-pancreas axis in liver-specific SMN-depleted mice.

2.7 Liver-specific SMN-depleted mice have normal survival rates and motor function

We also conducted a comprehensive assessment of survival, weight, and motor function in *Alb*^{Cre/+};*Smn*^{2B/F7} and *+/+;**Smn*^{2B/F7} mice up to postnatal day 60 (**Fig. 9A**). While the *Smn*^{2B/-} mice typically have a mean survival of 25 days²⁵, both *Alb*^{Cre/+};*Smn*^{2B/F7} and *+/+;**Smn*^{2B/F7} mice in our study survived up to 60 days (**Fig. 9B**), after which they were euthanized for tissue collection. Moreover, throughout the observation period, mice from both groups exhibited similar weight gain trends (**Fig. 9C**). When weight gain at P60 was normalized to the basal weight at P7, no statistically significant differences were observed (**Fig. 9D**).

To further evaluate motor function at different stages, we implemented several age-specific behavioral tests commonly used to assess motor function⁴⁰ (**Fig. 9E-G**). We previously showed *Smn*^{2B/-} mice have reduced motor function scores, as seen through increased time to right, prolonged balancing time and diminished muscle strength²⁶. Here, we found no alterations in *Alb*^{Cre/+};*Smn*^{2B/F7} compared to *+/+;**Smn*^{2B/F7} mice in time to right, as assessed by the righting reflex (**Fig. 9E**), nor in distal muscle strength measured through the inverted mesh grip test (**Fig. 9F**), or motor balance and coordination evaluated by the pen test (**Fig. 9G**). Overall, these findings collectively demonstrate that liver-specific SMN depletion does not impact survival, weight gain, or motor function.

3. Discussion

The liver is pivotal in regulating numerous biological functions in the body. These encompass, among others, the storage and balance of lipids, carbohydrates, and iron, as well as the synthesis of vital growth factors and the clearance of non-essential or harmful metabolites¹⁹. Given its multifaceted role, any alterations in liver function can potentially have cascading effects on other organs within the body. Hence, aside from elucidating the impact of liver-specific SMN depletion on hepatic function, it is imperative to explore its effects on other tissues, as alterations in these tissues have been documented in various mouse models of SMA^{11,18,34,41}.

Previous work on conditional knockout of *Smn* in the livers of mice resulted in severe impairment of liver development characterized by iron overload and subsequent liver atrophy, ultimately leading to late embryonic lethality⁸. In our study, we generated a novel mouse model where instead of a complete knockout of *Smn* in the liver, we created a scenario where SMN protein was depleted to 30% of full-length SMN production specifically within the liver, while SMN protein levels in other tissues remain comparable to those of heterozygous controls. Notably, this mouse line did not exhibit any signs of developmental delays, liver atrophy, or reduced lifespan. Nonetheless, within 19 days after birth, *Alb*^{Cre/+};*Smn*^{2B/F7} mice showed a mild yet rapid onset of fatty liver disease and trends towards an increase in lipid levels when maintained on a normal chow diet, underscoring the importance of SMN in lipid metabolism. Interestingly, these changes were transient, as by P60, liver-specific SMN-depleted mice displayed no lipid accumulation or hepatocyte morphological changes, likely due to the compensatory proliferation of non-recombined cells.

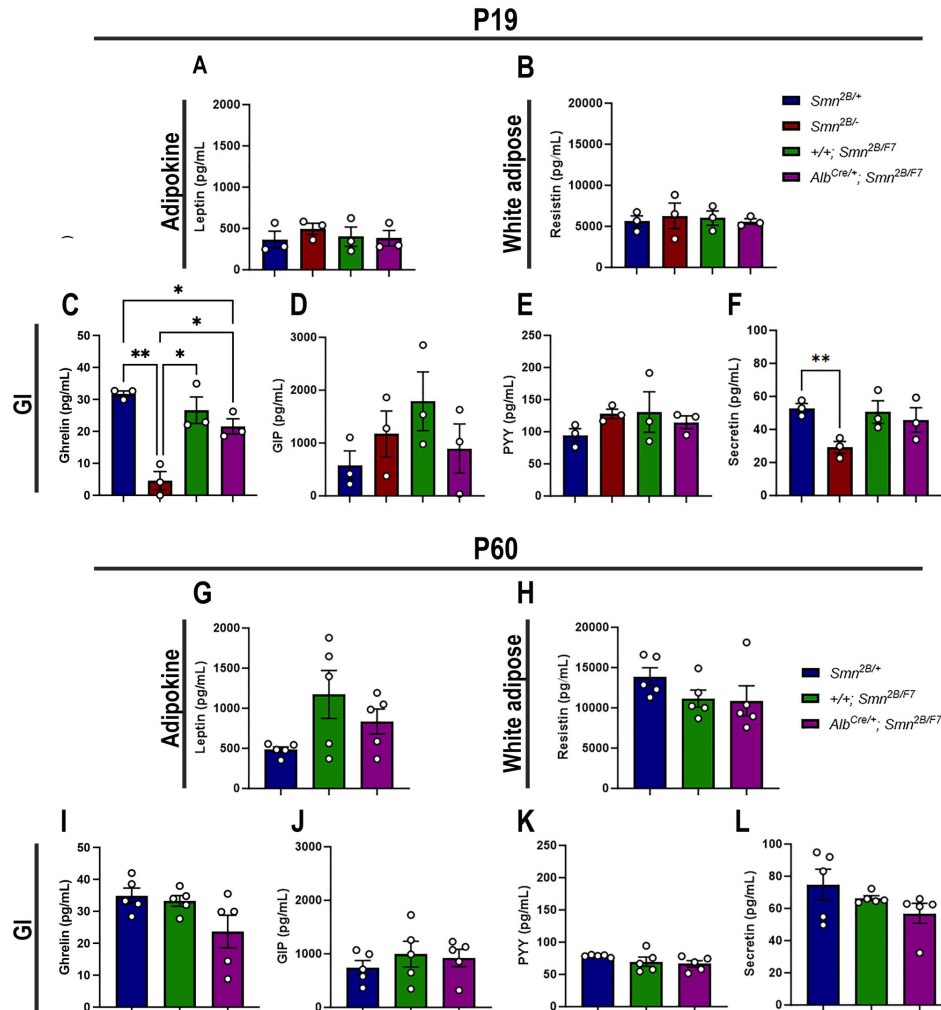


Figure 8.

Contribution of liver-specific SMN depletion to major adipocyte and gastrointestinal hormones at P19 and P60.

Panels A-B and G-H show bar graphs depicting non-fasting plasma levels of adipocyte-secreted hormones leptin (A, G) and resistin (B, H) in P19 (A-B) and P60 (G-H) mice across different genotypes. Panels C-F and I-L illustrate non-fasting plasma levels of gastrointestinal-secreted hormones, including ghrelin (C, I), gastric inhibitory polypeptide (D, J), peptide YY (E, K), and secretin (F, L) in P19 (C-F) and P60 (I-L) mice. $n \geq 3$, mean $\pm$ SEM. Statistical significance is indicated by * $p < 0.05$, ** $p < 0.01$, following Brown-Forsythe and Welch ANOVA.

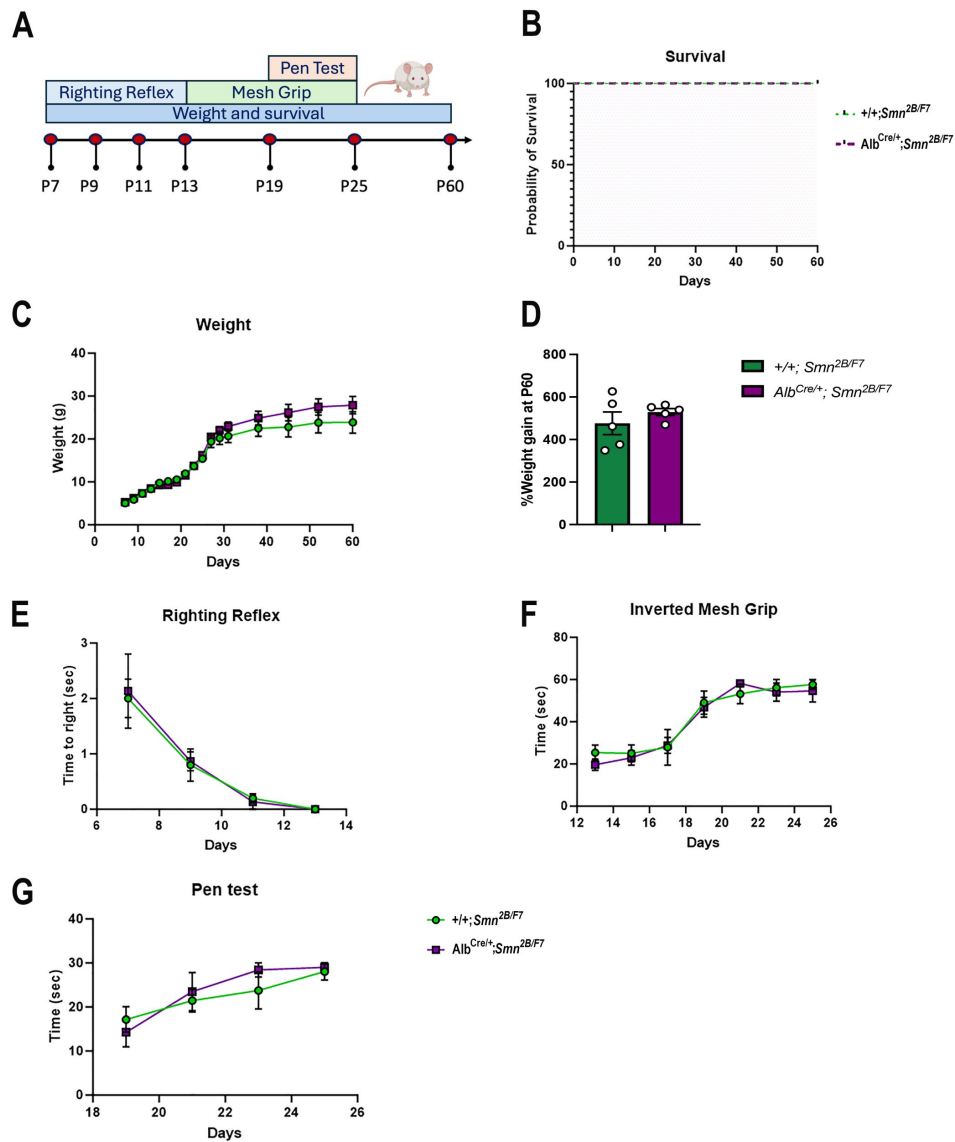


Figure 9.

Impact of liver-specific SMN depletion on motor function.

A. Schematic representation of experimental design. The righting reflex test was conducted from P7 to P13, the inverted mesh grip test from P13 to P25, and the pen test from P19 to P25. Assessments were performed every two days. Weight was measured every two days until day 30, and then weekly until day 60. Animal welfare was monitored throughout the 60-day period during weight measurements. B. Kaplan-Meier survival analysis; C, E-G: two-way ANOVA, followed by Šidák's method. D. Percentage weight gain at P60, normalized to basal weight at P7. n = 5 per genotype, mean ± SEM, two-tailed unpaired Student's t-test.

A recent study found that 75% of pediatric and adult SMA patients showed ultrasonic evidence of mild to moderate hepatic steatosis, independent of disease severity or SMA subtype⁴². In an *in vitro* model using SMA patient-derived induced pluripotent stem cell-derived hepatocyte-like cells (iHeps), a 10-fold increase in lipid accumulation was observed, correlating with reduced SMN protein expression. Proteomic and transcriptomic analyses revealed dysregulation in mitochondrial and lipid metabolism pathways, including downregulation of key genes involved in oxidative phosphorylation and fatty acid oxidation, alongside increased lipid biosynthesis. Notably, repletion of SMN in SMA iHeps restored metabolic and hepatic functions, rescuing steatosis and mitochondrial defects⁴². Excitingly, our current work further supports these findings, demonstrating that treatment with an adeno-associated viral vector carrying a liver-specific albumin promoter in *Smn*^{2B/-} mice resulted in partial restoration of liver SMN levels, improving survival, alleviating liver pathology, and restoring muscle size and pancreatic cell balance⁴³. Additionally, a recent study found that 60.8% of adult SMA type 3 patients (23 patients) exhibited at least one lipid abnormality⁴⁴, further supporting the hypothesis that hepatocyte-specific SMN deficiency contributes to liver pathology.

Mild liver steatosis may be accompanied by disruptions in crucial liver function markers, notably those implicated in iron metabolism, autophagy, and growth factor regulation, such as IGF-1¹⁸. Accordingly, we initially investigated the expression of HO-1, an enzyme responsible for heme degradation into iron, carbon monoxide, and biliverdin; and transferrin, a liver-synthesized glycoprotein involved in iron transport across tissues (reviewed in⁴⁵). Prior studies have documented alterations in these proteins in *Smn*^{2B/-} mouse livers¹⁸. While our current analysis did not reveal significant differences in *Alb*^{Cre/+}; *Smn*^{2B/F7} mice, the observed upward trend in transferrin and HO levels suggests ongoing changes in iron metabolism, which may not be fully manifested at P19. In addition, incomplete Cre-mediated excision of *Smn*^{F7} allele may be hindering a comprehensive understanding of SMN's role in liver function.

Furthermore, no alterations were observed for P62 levels, a multifunctional scaffold protein implicated in autophagy, formation of hepatic inclusion bodies, and hepatocyte cell death (reviewed in⁴⁶) despite its increased presence in mouse models of SMA¹⁸. Similarly, there were no changes detected in IGF-1 protein levels, whose deficiency has been linked to liver pathology and non-alcoholic fatty liver disease (NAFLD)^{47,48}, and its decrease is evident in *Smn*^{2B/-} mice¹⁸. NAFLD involves the accumulation of fatty acids in more than 5% of liver cells, occurring in the absence of alcohol consumption. This condition presents with varying degrees of severity, with simple steatosis being the most common manifestation. Simple steatosis is characterized by the accumulation of fat in the liver with minimal impact on liver function^{49,50}. Consequently, our model holds promise as a valuable resource for investigating microvesicular steatohepatitis and NAFLD during the early postnatal period. Additionally, unlike the well-studied *Smn*^{2B/-} model of SMA, this liver-specific SMN depletion enables the study of hepatic SMN function without the confounding effects of systemic SMN depletion or the severe phenotype observed in the *Smn*^{2B/-} mice.

Previously, our research unveiled metabolic abnormalities in SMA-like mice, including glucose intolerance, insulin hypersensitivity, and hyperglucagonemia, accompanied by changes in pancreatic islet composition characterized by an increase in glucagon-producing α -cells at the expense of insulin-producing β -cells¹¹. Histopathological examination of pancreatic tissue from infants and children with SMA type I corroborated these findings, revealing a predominant presence of glucagon-producing α -cells within pancreatic islets¹¹. Furthermore, one-year-old *Smn*^{+/-} mice producing approximately 50% of SMN protein, when exposed to a high-fat diet, exhibited dysregulation in the proportion of glucagon-producing α -cells within pancreatic islets and heightened hepatic insulin and glucagon sensitivity⁹.

Here, we uncovered a novel intrinsic role of liver-specific SMN in pancreatic function, distinct from IGF-1 signaling, pancreatic SMN levels (which remained stable), or canonical SMA pathology. Although the precise role of SMN in the liver-pancreas axis remains unclear, we speculate that altered amino acid metabolism may contribute to the disrupted communication between these two organs. The liver plays a central role in amino acid metabolism, and disturbances in this process can affect the liver- α cell axis, a well-documented feedback loop in which circulating amino acids stimulate glucagon secretion, which in turn regulates hepatic amino acid uptake and metabolism⁵¹. In the context of metabolic diseases such as NAFLD and SMA, liver dysfunction, including impaired lipid metabolism, may disrupt this axis, leading to abnormal glucagon secretion^{51,52}. It is possible that dysregulated amino acid pathways may directly influence this feedback loop.

Further evidence for this hypothesis comes from a recent study in individuals with type 2 diabetes (T2D) and NAFLD, which found hyperglucagonaemia and altered amino acid profiles. In T2D-NAFLD patients, plasma glucagon levels were significantly higher than in individuals without NAFLD, with a positive correlation between plasma glucagon concentrations and liver fat content. This study also showed increased plasma levels of isoleucine and glutamate, and decreased levels of glycine and serine in individuals with NAFLD⁵³. These findings suggest that metabolic dysregulation in the liver, including alterations in amino acid metabolism, may exacerbate glucagon secretion. Interestingly, such changes in amino acid metabolism are consistent with clinical observations in SMA patients, where an “amino acid diet” — low in fats and rich in specific amino acids — has been shown to provide some clinical benefit⁵⁴. Given this, it is plausible that impaired liver metabolism in SMN-depleted mice alters the liver- α cell axis, leading to dysregulated glucagon and GLP-1 secretion. While this proposed link between liver steatosis, amino acid metabolism, and pancreatic dysfunction is speculative, we are actively investigating the mechanisms underlying this liver-pancreas interplay in mouse models of SMA.

In addition, while the hallmark pathological features of SMA include motor neuron loss, NMJ pathology, muscle atrophy, and consequent motor function alterations, these were not evident in the *Alb^{Cre/+};Smn^{2B/F7}* mouse model at P19. However, given that glucose metabolism defects are observed in various neurodegenerative and neuromuscular disorders, subsequent work is required to clarify the impact of liver-specific SMN depletion on pancreatic function. Although hepatic and pancreatic abnormalities likely occur independently of disease onset, given that SMN depletion in motor neurons and muscle alone is sufficient to induce a SMA-like phenotype^{23,55,56}, the potential influence of these anomalies on SMA progression warrants careful consideration.

Overall, our findings suggest that alterations in SMN production solely within the liver may suffice to induce pathological hepatic and pancreatic function, shedding light on a novel role of SMN in liver physiology.

4. Materials and methods

4.1 Mouse models

The *Smn^{2B}* mouse, which harbors the *Smn^{2B}* mutation resulting from the substitution of three nucleotides within the exon splicing enhancer of exon 7, was developed and maintained on a C57BL/6 background in our laboratory^{10,24}. The FVB.129(B6)-*Smn1^{tm1Jme}* (*Smn^{F7/F7}*) and B6.Cg-Speer6-ps1^{Tg(Alb-Cre)21Mgn/J} (*Alb^{Cre}*) mice were acquired from The Jackson Laboratory. Liver-specific SMN-depleted mice were obtained through successive crosses between *Alb^{Cre/+};Smn^{F7/+}* and *Smn^{2B/F7}* mice (see **Fig. 1**). Line validation was conducted via genotyping of DNA extracted from mouse ear biopsies using PCR (see **Table 1** for list of primers used). Both male and female mice were included in the studies. All mice were bred and housed in the University of Ottawa

Animal Facility under protocol OHRI-3343, adhering to guidelines established by the Canadian Council on Animal Care. Additionally, we employed *Smn*^{2B/-} mice (a severe SMA mouse model) and asymptomatic heterozygous *Smn*^{2B/+} mice as controls in our experiments.

4.2 Tissue harvesting and processing

Following euthanasia, liver, and *tibialis anterior* (TA) muscles underwent fixation in a 1:10 dilution of buffered formalin for 48 h at 4°C, followed by transfer to 70% ethanol at the same temperature until processing. The pancreas was fixed in 4% paraformaldehyde (PFA) for 48 h at 4°C, then similarly moved to 70% ethanol for processing. For the lumbar spinal cord (SC) and liver (for Oil Red-O staining), fixation was performed overnight at 4°C in 4% PFA, followed by immersion in 30% sucrose in PBS. Subsequently, the liver and spinal cord were flash-frozen in optimal cutting temperature (OCT) embedding medium. Cryosections of the lumbar spinal cord were cut at a thickness of 16 µm, while liver sections were cut at 10 µm and stored at -80°C until immunohistochemical analysis. Cre immunostaining of cryosectioned liver and spinal cord was performed as described in [section 4.4.1](#), with the exception that the antigen retrieval step was not performed.

The abdominal musculature was dissected and fixed in 4% PFA for 15 min at room temperature, followed by 3 washes in 1X PBS, after which the transversus abdominis (TVA) muscle was dissected from the abdominal musculature. TA, liver, and pancreas samples were handled at the University of Ottawa Department of Pathology and Laboratory Medicine. Paraffin blocks containing tissues were sectioned at 4 µm thickness using a microtome. Sections of TA and liver were stained with hematoxylin and eosin (H&E) using an XL CV5030 autostainer from Leica. Liver cryosections were stained with Oil Red-O and counterstained with hematoxylin using established protocols. Images of H&E and Oil Red-O-stained samples were scanned with a MIRAX MIDI digital slide scanner manufactured by Zeiss. Image acquisition was performed using Panoramic Viewer 1.15.4 software.

4.3 Immunoblotting

Total protein was extracted by homogenizing flash-frozen samples of spinal cord, brain, liver, pancreas and hindlimb muscles using RIPA lysis buffer (Cell Signaling), supplemented with phenylmethylsulfonyl fluoride (PMSF, Cell Signaling). The concentrations of the extracted proteins were determined using the Pierce BCA Protein Assay kit (ThermoFisher).

For liver, brain, and spinal cord tissues, 20 µg of protein per sample were separated by 10% sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE), while 30 µg were used for pancreas and muscle samples. Following electrophoresis, the proteins were transferred onto a PVDF membrane (Immobilon-P, Millipore) and blocked for 30 min at room temperature using intercept blocking buffer (LI-COR). Subsequently, the membranes were incubated overnight at 4°C with primary antibodies.

After incubation, the membranes were rinsed with tris-buffered saline containing 0.1% Tween-20 (TBS-T) and then incubated for 1 h at room temperature with IRDye 680R or 800CW (LI-COR) secondary antibody. Fluorescent signals were detected using the Odyssey Infrared Imaging System (LI-COR). Total raw values were normalized to α-tubulin (housekeeping protein) obtained from the same blot.

4.4 Immunohistochemistry

NAME	SEQUENCE (5'-3')
Alb-Cre	TGCAAACATCACATGCACAC
	TTGGCCCCTTACCATAACTG
	GAAGCAGAAGCTTAGGAAGATGG
SnnF7	AGAAGGAAAGTGCTCACATACAAATT
	TGTCTATAATCCTCATGCTATGGAG
Snn2B	GGTTTCAGACAAAATAAAAAAGAATTTAAG
	TTTGGCAGACTTTAGCAGGGC

Table 1.

List of primers used for genotyping

4.4.1 Pancreas

Pancreatic sections underwent deparaffinization using three washes of Histo-Clear (National Diagnostics, Atlanta, GA) for 5 min each, followed by two washes of a 50/50 mixture of absolute ethanol and Histo-Clear for 3 min each. Subsequently, the slides were gradually rehydrated in two washes of 100% (v/v) ethanol (3 min each), followed by one wash of 95% (v/v), 70% (v/v), 50% (v/v), and 0% (v/v) ethanol for 3 min each. The slides were then incubated in 0.5% Triton-X-100 (Millipore Sigma, Burlington, MA) in PBS for 5 min, followed by three washes with PBS. Afterward, the slides were blocked in a solution containing 20% goat serum and 0.3% Triton-X-100 in TBS for 2 h.

Primary antibodies targeting insulin, glucagon or Cre ([Table 2](#)) were applied to the slides in a solution containing 2% goat serum and 0.3% Triton-X-100 in TBS overnight at 4°C. The slides were then washed three times with PBS and subsequently incubated with secondary antibodies in a solution containing 2% goat serum and 0.3% Triton-X-100 for 1 h. Following this, the solution was removed, and DAPI (1:1,000) in PBS was added for 5 min at room temperature. The slides were washed three times with PBS and mounted in Fluoromount-G™ Mounting Medium (Invitrogen).

Images were taken using an Axio Imager M2 microscope (Zeiss) and captured with a 40X objective. The number of glucagon and insulin positive cells per islet was counted for each mouse.

4.4.2 Motor neurons

Lumbar spinal cord sections were prepared for choline acetyltransferase (ChAT) staining of motor neurons. The slides were air-dried at room temperature for 30 min and then rinsed in TBS-T for 5 min. Samples were permeabilized in 0.3% Triton X-100 in PBS for 30 min, followed by blocking in 1x Power Block (BioGenex, Fremont, CA) for 20 min at room temperature. Subsequently, the samples were incubated with a goat anti-ChAT (details in [Table 2](#)) antibody at a dilution of 1:100 in 1% BSA and 0.3% Triton X-100 in PBS for 3 nights at 4°C.

After the initial antibody incubation, slides were washed twice for 10 min with TBS-T at room temperature. Following the washes, samples were then incubated with Alexa Fluor 555 donkey anti-goat IgG at a dilution of 1:200 in 10% donkey serum and 0.3% Triton X-100 in PBS for 2 h at room temperature in a humid chamber. Nuclei were counterstained with 4',6-diamidino-2-phenylindole (DAPI) at a dilution of 1:1000 in PBS for 5 min.

Finally, the slides were rinsed 3X for 10 min with TBS-T and mounted with Fluoromount-G™ Mounting Medium (Invitrogen). Spinal cord sections were examined under fluorescence using an Axio Imager M2 microscope (Zeiss) and imaged at 20X magnification for quantification, and at 63x for representative pictures. The number of ChAT-positive motor neurons with a diameter of ≥ 20 μ m per ventral horn was recorded for 7-10 different sections per animal, each separated by at least 100 μ m to prevent re-counting of the same motor neuron. An average number of motor neuron cell bodies per section was determined.

4.4.3 Neuromuscular Junction (NMJ)

To evaluate neuromuscular innervation, we performed immunohistochemistry using established protocols²⁶. After euthanasia, the TVA muscles were promptly dissected and fixed in 4% paraformaldehyde (Electron Microscopy Science) in PBS for 15 min. Post-synaptic acetylcholine receptors (AChRs) were labeled for 10 min with alpha-bungarotoxin (aBTX) conjugated to Alexa Fluor 488. Subsequently, the muscles were blocked in 4% bovine serum albumin (BSA) and 1% Triton-X-100 in PBS for 30 min.

Following blocking, the muscles underwent overnight incubation with primary antibodies for neurofilament and synaptic vesicle protein 2 (refer to [Table 2](#)). Visualization of the labeled structures was achieved using DyLight-conjugated secondary antibodies. The muscles were then whole-mounted in Dako Fluorescent mounting media and imaged with a 20x objective on a Zeiss Axio Imager M1 microscope.

A minimum of three fields of view (FOV) were quantified per muscle. For each FOV, the percentage of fully occupied endplates was noted. Fully occupied endplates were defined as motor endplates completely covered by the presynaptic terminal labeled with synaptic vesicle protein 2 (SV2) and neurofilament (NF).

4.5 Muscle fiber analysis

Muscle fiber analysis was performed using ImageJ software (version 2.9.0/1.53t). A total of one hundred fibers were evaluated per animal, covering various regions of the muscle section to ensure good representation. The area of each fiber was measured in square micrometers (μm^2) to compute both the average and distribution of fiber sizes for each animal.

4.6 Glucose levels

Non-fasting blood glucose concentrations were assessed using a OneTouch Ultra2 Blood Glucose Test Strips (LifeScan Europe GmbH, Zug, Switzerland) immediately following blood collection. Approximately 2 μL of blood was applied to the test strip for glucose concentration measurement.

4.7 ELISA for Insulin-like Growth Factor I (IGF-1)

The concentration of IGF-1 in the liver was quantified using the Mouse/Rat IGF-I/IGF1 Quantikine ELISA Kit (MG100, R&D Systems, Inc., Minneapolis, MN, USA). Liver protein lysates were diluted at a ratio of 1:6 in calibrator diluent and the assay was performed following the manufacturer's instructions.

4.8 Triglycerides quantification

Liver tissues from P19 mice were promptly dissected and flash frozen. Subsequently, triglyceride analysis was conducted at the Vanderbilt Mouse Metabolic Phenotyping Center, employing established protocols previously utilized by our team¹⁸.

4.9 Mouse metabolic hormone assay

Blood was collected from the mice immediately after decapitation using a Microcuvette CB 300 K2E coated with K2 EDTA (16.444.100) for capillary sampling. All samples were taken randomly without a fasting period. The blood was centrifuged at 5000 g for 5 minutes at room temperature to separate the plasma. For plasma volumes less than 50 μL , genotypes were pooled to achieve a minimum volume of 50 μL as needed. The multiplexing analysis was performed using the Luminex™ 200 system (Luminex, Austin, TX, USA) by Eve Technologies Corp. (Calgary, Alberta). Twelve markers were simultaneously measured in the samples using Eve Technologies' Mouse Metabolic Hormone 12-Plex Discovery Assay® (MilliporeSigma, Burlington, Massachusetts, USA) according to the manufacturer's protocol. The 12-plex consisted of Amylin(active), C-Peptide 2, Ghrelin, GIP(total), GLP-1(active), Glucagon, Insulin, Leptin, PP, PYY, Resistin and Secretin. Assay sensitivities of these markers range from 1.4 – 91.8 pg/mL for the 12-plex. Individual analyte sensitivity values are available in the MilliporeSigma MILLIPLEX® MAP protocol. If analyte levels were too low to be detected and fell outside the dynamic range, they were recorded as 0 in the graphs.

Table 2.

List of antibodies used.

<i>Application</i>	<i>Antibody</i>	<i>Species</i>	<i>Dilution</i>	<i>Company (Catalog #)</i>
Western Blot	SMN	Mouse	1:2 500	BD Transduction (610647)
Western Blot	Alpha-tubulin	Rabbit	1:10 000	Abcam (ab4074)
Western Blot	Alpha-tubulin	Mouse	1:10 000	Sigma-Aldrich (CP06)
Western Blot	Heme oxygenase-1 (HO)	Rabbit	1:10 000	Abcam (ab68477)
Western Blot	P62	Mouse	1:1 000	Abcam (ab56416)
Western Blot	Transferrin	Rabbit	1:1 000	Abcam (ab82411)
Western Blot	Goat anti-mouse IR Dye 800 CW		1:5 000	LI-COR (925-32210)
Western Blot	Goat anti-mouse IR Dye 680 R		1:5 000	LI-COR (926-68070)
Western Blot	Goat anti-rabbit IR Dye 800 CW		1:5 000	LI-COR (925-32211)
Western Blot	Goat anti-rabbit IR Dye 680 R		1:5 000	LI-COR (926-68071)
IHC	ChAT	Goat	1:100	Millipore (AB144P)
IHC	Donkey anti-goat Alexa Fluor 555		1:200	Invitrogen (A21432)
IHC	TRITC conjugated bungarotoxin	N/A	1:1 000	Invitrogen (T1175)
IHC	Neurofilament (NF-M)	Mouse	1:100	(Developmental Studies Hybridoma Bank, P12839)
IHC	Synaptic vesicle glycoprotein 2A (SV2A)	Mouse	1:250	(Developmental Studies Hybridoma Bank, Q7L0J3)
IHC	Goat anti-mouse Alexa Fluor 488		1:250	Invitrogen (A11001)
IHC	Glucagon	Mouse	1:200	Abcam (ab10988)
IHC (Parafin)	Cre Recombinase (D7L7L)	Rabbit	1:500	Cell Signaling (15036)
IHC (Frozen)			1:200	
IHC	Insulin	Rabbit	1:50	Abcam (ab181547)
IHC	Goat anti-mouse Alexa Fluor 555		1:500	Invitrogen (A21422)
IHC	Goat anti-rabbit Alexa Fluor 488		1:500	Invitrogen (A11034)

4.10 Motor function, weight, and survival assessments

Motor function, weight, and survival were assessed according to established protocols⁴⁰. Spinal reflexes were evaluated using the righting reflex test, while motor balance and coordination were evaluated with the pen test. Additionally, muscle strength was measured using the inverted mesh grip test (see Fig. 9A). The righting reflex test was conducted from P7 to P13, the inverted mesh grip test from P13 to P25, and the pen test from P19 to P25. Assessments were performed every two days. A maximum duration of 30 sec for the pen test and 60 sec for the inverted mesh grip represented the established threshold. Three consecutive measurements were obtained, and their average was recorded for subsequent analysis. Weight was measured every two days starting at day 7 until day 30, and then weekly until day 60. Animal welfare was monitored throughout the 60-day period during weight measurements.

4.11 Statistical analysis

Survival data were visualized using Kaplan-Meier survival curves, and intergroup differences were evaluated using the Mantel-Cox test. Brown-Forsythe and Welch ANOVA were employed to compare three or more sets of unpaired measurements, given the mosaic-like outcome observed in the *Alb^{Cre/+};Smn^{2B/F7}* model, where assumptions of equal variances were not tenable. Two-way ANOVA, followed by Šidák's method for multiple comparisons, was utilized to assess the influence of two factors on a response. Pairwise comparisons were performed using a two-tailed Student's t-test. Statistical analyses were conducted using GraphPad Prism V.10.2.1 (GraphPad Prism Software, San Diego, CA), with significance set at $P < 0.05$. Data were presented as mean $\pm$ standard error of the mean. Sample sizes (n) are provided in figure legends, indicating the number of biological replicates analyzed, with each data point corresponding to an individual mouse from at least two different litters. Detailed statistical information is outlined in the respective figure legends, where statistical significance is indicated by *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

Data sharing

All authors had access to the study data and reviewed and approved the final manuscript. All data associated with this study are available in the main text or supplementary materials. Raw data can be provided upon request.

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Additional information

Declaration of interests

The authors disclose no conflicts of interest. The funders mentioned above were not involved in the study design, data collection and analysis, decision to publish, or manuscript preparation.

Author's contributions

M.M.A.A. conceived and designed the study, with supervision and support from R.K. The methodology was developed by M.M.A.A., Y.D.R., and S.G. Formal analysis was conducted by M.M.A.A. and E.R.S. The investigation was carried out by M.M.A.A. The manuscript was written by M.M.A.A and critically revised by R.K. Funding acquisition was managed by R.K.

Additional files

Supplementary Figure 1 [↗](#)

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Reviewer #1 (Public review):

Summary:

This manuscript presents a comprehensive exploration of the role of liver-specific Survival Motor Neuron (SMN) depletion in peripheral and central nervous system tissue pathology through a well-constructed mouse model. This study is pioneering in its approach, focusing on the broader physiological implications of SMN, which has traditionally been associated predominantly with spinal muscular atrophy (SMA).

Strengths:

- (1) Novelty and Relevance: The study addresses a significant gap in understanding the role of liver-specific SMN depletion in the context of SMA. This is a novel approach that adds valuable insights into the multi-organ impact of SMN deficiency.
- (2) Comprehensive Methodology: The use of a well-characterized mouse model with liver-specific SMN depletion is a strength. The study employs a robust set of techniques, including genetic engineering, histological analysis, and various biochemical assays.
- (3) Detailed Analysis: The manuscript provides a thorough analysis of liver pathology and its potential systemic effects, particularly on the pancreas and glucose metabolism.
- (4) Clear Presentation: The manuscript is well written. The results are presented clearly with well-designed figures and detailed legends.

Weaknesses:

(1) Limited Time Points: The study primarily focuses on a single time point (P19). This limits the understanding of the temporal progression of liver and pancreatic pathology in the context of SMN depletion. Longitudinal studies would provide a better understanding of disease progression.

(2) Incomplete Recombination: The mosaic pattern of Cre-mediated excision leads to variability in SMN depletion, which complicates the interpretation of some results. Ensuring more consistent recombination across samples would strengthen the conclusions.

After the revision, the authors addressed the reviewers' questions by extending their analyses to include P60 mice, conducting both liver and pancreatic analyses, and adding a comprehensive panel of metabolic hormones related to glucose metabolism in animals at P19 and P60. They also corrected all errors identified during the initial review process and expanded the discussion to clarify raised issues. All my questions have now been addressed.

<https://doi.org/10.7554/eLife.99141.2.sa2>

Reviewer #2 (Public review):

Summary:

Marylin Alves de Almeida et al. developed a novel mouse cross via conditionally depleting functional SMN protein in the liver (AlbCre/+;Smn2B/F7). This mouse model retains a proportion of SMN in the liver, which better recapitulates SMN deficiency observed in SMA patients and allows further investigation into liver-specific SMN deficiency and its systemic impact. They show that AlbCre/+;Smn2B/F7 mice do not develop an apparent SMA phenotype as mice did not develop motor neuron death, neuromuscular pathology or muscle atrophy, which is observed in the Smn2B/- controls. Nonetheless, at P19 and P60, these mice develop mild liver steatosis, and interestingly, this conditional depletion of SMN in the liver impacts cells in the pancreas.

Strengths:

The current model has clearly delineated the apparent metabolic perturbations which involve a significantly increased lipid accumulation in the liver and pancreatic cell defects in AlbCre/+;Smn2B/F7 mice at P19 and P60. Standard methods like H&E and Oil Red-O staining show that in AlbCre/+;Smn2B/F7 mice, their livers closely mimic the livers of Smn2B/- mice, which have the full body knockout of SMN protein. Unlike previous work, this liver-specific conditional depletion of SMN is superior in that it is not lethal to the mouse, which allows an opportunity to investigate the long-term effects of liver-specific SMN on the pathology of SMA.

Weaknesses:

Given that SMA often involves fatty liver, dyslipidemia and insulin resistance, using the current mouse model, the authors could have explored the long-term effects of liver-specific depletion of SMN on metabolic phenotypes beyond P19, as well as systemic effects like glucose homeostasis. Given that the authors also report pancreatic cell defects, the long-term effect on insulin secretion and resistance could be further explored. This has been addressed in the revised manuscript. The mechanistic link between a liver-specific SMN depletion and apparent pancreatic cell defects has been made clearer.

Discussion:

This current work explores a novel mouse cross in order to specifically deplete liver SMN using an Albumin-Cre driver line. This provides insight into the contribution of liver-specific SMN protein to the pathology of SMA, which is relevant for understanding metabolic perturbations in SMA patients. Nonetheless, given that SMA in patients involve a systemic deletion or mutation of the SMN gene, the authors could emphasize the utility of this liver-specific mouse model, as opposed to using in vitro models, which have been recently reported (Leow et al, 2024, JCI).

Comments on current version:

No further suggestions. Previous recommendations have been addressed by the authors.

<https://doi.org/10.7554/eLife.99141.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This manuscript presents a comprehensive exploration of the role of liver-specific Survival Motor Neuron (SMN) depletion in peripheral and central nervous system tissue pathology through a well-constructed mouse model. This study is pioneering in its approach, focusing on the broader physiological implications of SMN, which has traditionally been associated predominantly with spinal muscular atrophy (SMA).

Strengths:

(1) Novelty and Relevance: The study addresses a significant gap in understanding the role of liver-specific SMN depletion in the context of SMA. This is a novel approach that adds valuable insights into the multi-organ impact of SMN deficiency.

(2) Comprehensive Methodology: The use of a well-characterized mouse model with liver-specific SMN depletion is a strength. The study employs a robust set of techniques, including genetic engineering, histological analysis, and various biochemical assays.

(3) Detailed Analysis: The manuscript provides a thorough analysis of liver pathology and its potential systemic effects, particularly on the pancreas and glucose metabolism.

(4) Clear Presentation: The manuscript is well written. The results are presented clearly with well-designed figures and detailed legends.

We thank the reviewer for their positive comments. They had some concerns for us to consider (see below). We provide a point-by-point response to their comments.

Weaknesses:

(1) Limited Time Points: The study primarily focuses on a single time point (P19). This limits the understanding of the temporal progression of liver and pancreatic pathology in the context of SMN depletion. Longitudinal studies would provide a better understanding of disease progression.

We thank the reviewer for the suggestion. We extended our analysis to include P60 mice and performed both liver and pancreatic analyses at this time point to address this suggestion.

(2) Incomplete Recombination: The mosaic pattern of Cre-mediated excision leads to variability in SMN depletion, which complicates the interpretation of some results. Ensuring more consistent recombination across samples would strengthen the conclusions.

The variability in Cre-mediated excision is inherently stochastic, influenced by factors such as Cre expression levels, timing of recombination, and the accessibility of the target locus in individual cells. Achieving complete consistency across samples is particularly challenging, especially given the complexity of our breeding scheme, which occasionally results in litters without any animals of the desired genotype. Importantly, our study not only demonstrates that liver-specific SMN depletion results in liver alterations and pancreatic dysfunction but also highlights the limitations and challenges associated with this mouse model. By doing so, we aim to provide valuable insights for other researchers considering similar approaches in future studies.

Reviewer #2 (Public review):

Summary:

Marylin Alves de Almeida et al. developed a novel mouse cross via conditionally depleting functional SMN protein in the liver (AlbCre/+;Smn2B/F7). This mouse model retains a proportion of SMN in the liver, which better recapitulates SMN deficiency observed in SMA patients and allows further investigation into liver-specific SMN deficiency and its systemic impact. They show that AlbCre/+;Smn2B/F7 mice do not develop an apparent SMA phenotype as mice did not develop motor neuron death, neuromuscular pathology or muscle atrophy, which is observed in the Smn2B/- controls. Nonetheless, at P19, these mice develop mild liver steatosis, and interestingly, this conditional depletion of SMN in the liver impacts cells in the pancreas.

Strengths:

The current model has clearly delineated the apparent metabolic perturbations which involve a significantly increased lipid accumulation in the liver and pancreatic cell defects in AlbCre/+;Smn2B/F7 mice at P19. Standard methods like H&E and Oil Red-O staining show that in AlbCre/+;Smn2B/F7 mice, their livers closely mimic the livers of Smn2B/- mice, which have the full body knockout of SMN protein. Unlike previous work, this liver-specific conditional depletion of SMN is superior in that it is not lethal to the mouse, which allows an opportunity to investigate the long-term effects of liver-specific SMN on the pathology of SMA.

We thank the reviewer for their positive comments. They had some concerns for us to consider (see below). We provide a point-by-point response to their comments (review comments in black, our response in red).

Weaknesses:

Given that SMA often involves fatty liver, dyslipidemia and insulin resistance, using the current mouse model, the authors could have explored the long-term effects of liver-specific depletion of SMN on metabolic phenotypes beyond P19, as well as systemic effects like glucose homeostasis. Given that the authors also report pancreatic cell defects, the long-term effect on insulin secretion and resistance could be further

explored. The mechanistic link between a liver-specific SMN depletion and apparent pancreatic cell defects is also unclear.

We extended our analysis to include P60 mice and performed both liver and pancreatic analyses at this time point to address this suggestion. In addition, we discussed the liver-pancreas axis in the Discussion.

Discussion:

This current work explores a novel mouse cross in order to specifically deplete liver SMN using an Albumin-Cre driver line. This provides insight into the contribution of liver-specific SMN protein to the pathology of SMA, which is relevant for understanding metabolic perturbations in SMA patients. Nonetheless, given that SMA in patients involve a systemic deletion or mutation of the SMN gene, the authors could emphasize the utility of this liver-specific mouse model, as opposed to using in vitro models, which have been recently reported (Leow et al, 2024, JCI). Authors should also discuss why a mild metabolic phenotype is observed in this current mouse model, as opposed to other SMA mouse models described in literature.

We appreciate the reviewer's insightful comment. We have thoroughly addressed this suggestion in the Discussion section, particularly in lines 284-298; 309-322 and 334-359.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Longitudinal Studies: Conducting studies at maybe one more time points postnatally to provide a clearer picture of how liver-specific SMN depletion affects tissue pathology over time.

We extended our analysis to include P60 mice and performed both liver and pancreatic analyses at this time point to address this suggestion.

(2) Functional Assays: Incorporate glucose tolerance tests, insulin sensitivity tests, and more detailed metabolic profiling to better understand the physiological consequences of liver-specific SMN depletion on glucose metabolism and pancreatic function.

We sincerely thank the reviewer for this suggestion. We have included a full panel of metabolic hormones associated with glucose metabolism from animals at P19 and P60. These new data, along with additional figures, have now been provided in our revised manuscript.

(3) Mechanism: Discuss the molecular pathways affected by SMN depletion in the liver and pancreas. Mechanistic studies including transcriptomic or proteomic analyses to identify dysregulated pathways will help.

We appreciate the reviewer's insightful comment. We have thoroughly addressed this suggestion in the Discussion section, particularly in lines 284-298 and 334-359.

(4) Typos in the abstract: beta cells secrete insulin and alpha cells produce glucagon.

Thank you for catching this error. It has been corrected to reflect that insulin is produced by beta cells and glucagon by alpha cells.

(5) Efficiency and specificity of the Alb-Cre: if possible, cross the Alb-Cre with the Rosa26 reporter line to test the efficiency and specificity of the Alb-Cre.

We agree that this would provide valuable insights. However, initiating a new breeding program to generate the required genotypes would take over a year and is beyond the scope of this study. To address this in part, we performed Cre immunostaining of the liver, pancreas, and spinal cord at P19, as well as the liver at P60. These results, now included in Supplemental Figure 1, demonstrate liver-specific expression and variability across hepatocytes.

Reviewer #2 (Recommendations for the authors):

The title of this manuscript is potentially misleading. The manuscript largely investigates the involvement of SMN protein on peripheral organs such as the liver, muscles, neuromuscular junction, and the pancreas. Yet, the title could be interpreted that the peripheral nervous system or central nervous system is the main focus. The title should be edited to indicate key terms such as "motor neuron and peripheral tissue pathology".

Thank you for pointing this out. We have revised the title to better represent the study's focus. It is now "Impact of liver-specific survival motor neuron (SMN) depletion on central nervous system and peripheral tissue pathology".

Suggestions:

Please clarify and explain clearly the various mouse lines (Smn2B/+, Smn2B/- and +/-; Smn2B/F7) used as controls as the nomenclature used is confusing. In addition, authors could consider the use of a wild-type mouse line to be used as a control to validate changes in AlbCre/+;Smn2B/F7 mice.

We have now provided clarification on mouse line nomenclature in the Results section (lines 104–124). Full-body heterozygous mice (Smn2B/+) are used as controls due to their slightly reduced SMN protein levels and absence of phenotypic changes compared to wild-type mice.

Given that the main phenotype implicated by the liver-specific depletion of SMN protein in AlbCre/+;Smn2B/F7 mice is pancreatic abnormalities (changes in alpha- and beta- cell numbers and blood glucose levels), authors should expand further on the pancreatic phenotype.

We added a full panel of metabolic hormones related to glucose metabolism in animals at P19 and P60. Furthermore, this has been discussed in detail in lines 284-298 and 334-344 of the Discussion.

A pancreas-specific depletion of SMN would provide this current manuscript with a better understanding of the role of SMN in regulating SMA pathology and provide more definitive conclusions on the contribution of liver-specific SMN depletion on normal pancreatic function.

We agree that this would be very informative. However, to do this would require initiation of a new breeding program that will take more than a year to arrive at the right genotypes. Although valuable, it is beyond the scope of the present study.

The authors should also delineate the role of hepatic SMN in pancreatic function, and how the intrinsic liver-specific loss of SMN directly impacts the pancreas. Currently, literature demonstrates that the fatty liver phenotype in SMA patients is a primary SMN-dependent hepatocyte-intrinsic liver defect associated with mitochondrial and other hepatic metabolism implications (see Leow et al, 2024 J Clin Invest). Given that the authors describe that SMN protein levels are not altered in the pancreas of

AlbCre/+;Smn2B/F7 mice at P19, the authors ought to clarify how pancreas development and function is impacted in this mouse model, whether in-utero or postnatally. This could potentially underscore the cross-talk between liver SMN and pancreas function.

We have discussed the relationship between hepatic SMN and pancreatic function in the Discussion at lines 284-298 and 334-359.

Authors should also perform some metabolic tolerance tests to both oral glucose and insulin at an older age (e.g. P60) to study their homeostasis in these mice. These would help to substantiate the authors' conclusion and provide the paper with a greater level of novelty.

We thank the reviewer for this suggestion. A full panel of metabolic hormones related to glucose metabolism at P19 and P60 has been included, supported by additional figures that enhance the manuscript's novelty and depth.

Authors mentioned in the Discussion in lines 238 to 240: "Altogether, our findings underscore the necessity of conducting further investigations at later time points to unveil potential modifications in other pathways and their repercussions on liver physiology". Please elucidate the effects of longer term liver-specific depletion of SMN beyond P19, such as the onset of NAFLD or a diabetic phenotype due to pancreatic dysfunctions.

We extended our data to include P60 mice and performed liver and pancreatic analyses at these time points. The observed effects were transient, possibly due to the stochastic nature of Cre expression.

In addition, while AlbCre/+;Smn2B/F7 mice had similar weight gain trends as controls, it does appear that AlbCre/+;Smn2B/F7 mice weigh more than their controls by P60 (Figure 9C). This data would provide more convincing evidence of the metabolic defects observed in these mice.

As per the reviewer's suggestion, we included new data (Figure 9D) showing % weight gain at P60 normalized to basal weight at P7. However, no statistically significant differences were detected.

Other than protein quantification, authors should perform immunohistochemistry or in-situ hybridization of SMN and imaging of AlbCre/+;Smn2B/F7 organs to validate the loss of liver-specific SMN. It is unclear from western blots that the expression of SMN is only in hepatocytes.

We thank the reviewer for the suggestion. Unfortunately, SMN antibodies have not produced reliable tissue immunostaining. To address this, we performed Cre immunostaining of the liver, pancreas, and spinal cord at P19, and the liver at P60, which demonstrated liver-specific expression. These results are now included in Supplemental Figure 1.

Authors should consider re-wording lines 228 through 231: "While our current analysis did not reveal significant differences in AlbCre/+;Smn2B/F7 mice, the observed upward trend in transferrin and HO levels suggests ongoing changes in iron metabolism, which may not be fully manifested at P19". Alternatively, a higher number of mouse samples would allow them to qualify this statement. Authors should also consider comparing levels of liver biomarkers such as ALT and AST, to check for liver homeostatic function.

We have removed speculative statements to avoid unsupported claims.

Recommendations:

The methods and additional details to generate the AlbCre^{+/+};Smn2B/F7 should be explained better in section 2.1 of the Results. It is potentially confusing as to why these mice had to carry both 2B and F7 alleles. Additionally, the role of the F7 allele is not deliberately clear in the Introduction.

Additional details are now included in the Introduction (lines 87-90) and the Results section (lines 104-124).

Authors should refer to Leow et al 2024 (J Clin Invest) and discuss how their current findings compare with their hepatocyte-intrinsic SMN deficiency IPSCs model. We note a previous publication (Deguise et al 2021 Cell Mol Gastroenterol Hepatol) by the authors which characterized the Smn2B^{-/-} mouse model and its NAFLD/NASH features. From our understanding, the Smn2B^{-/-} mouse model appears to recapitulate SMA phenotype well, such as the early onset of hepatic steatosis and neurological conditions. As a follow-up to this publication, authors should discuss why this current study of a liver-specific SMN depletion is important and relevant to the study of SMA pathology.

We thank the reviewer for the insightful suggestions. We have included a discussion of these findings and their relevance to the study of SMA pathology in lines 284-298 and 309-322.

Minor corrections:

Abstract (line 32) reads: "a decrease in insulin producing alpha-cells and an increase in glucagon producing beta-cells". The authors should clarify and correct as insulin producing beta-cells and glucagon producing alpha-cells.

Thank you for catching the error. We corrected the description of insulin- and glucagon-producing cells.

Please clarify the number and gender of mice used for weight tracking and motor function experiments up to P60 (Figure 9C). It would be inappropriate if male and female mice were plotted together. If so, authors should stratify data by gender.

We thank the reviewer for the suggestion. Unfortunately, we did not stratify the animals by sex due to the unequal and insufficient number of males and females in our study. To address this, we normalized weight gain to each animal's starting weight, and no significant differences were observed (now shown in Figure 9D).

The number of figures should be reduced. We recommend merging Figures 1 and 2 (generation of AlbCre^{+/+};Smn2B/F7 mouse line and validation) and Figures 3 and 4 (liver function). Figures 5 through 9 may be supplemental figures instead.

We thank the reviewer for the suggestions. We merged Figures 1 and 2, and Figures 3 and 4, as requested. However, we would prefer to keep the other figures within the main results as they assess the impact of liver-specific depletion of SMN on other pathologies within the mouse model.

Standardize the use of asterisks and reporting p-values in Figure 2. All other figures in the manuscript utilize asterisks, but Figures 2C', 2D' and 2E' use p-values across comparisons.

P-values were included only when they approached statistical significance, providing additional clarity to the results.

| *It is unclear what the white arrow in Figure 7A indicates.*

It is meant to point out the absence of an innervating axon. Please see Figure 5 legend, lines 801-802.

| *Note spelling errors in Figures 8B and 8C: 'Muscle flber'.*

Thank you for catching this. We have corrected the typo to indicate muscle fiber instead.

| *Please clarify if muscle fiber size should be indicated as μm^2 instead of μ^2 in Figures 8B and 8C, as written in Materials and Methods under line 394.*

Thank you for catching this. We corrected the typo to indicate μm^2 instead.

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